GENETICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: Newborn Screening

Grading guide. Total **24 points**. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — The chain

1. Fill in the chain for PKU, one link per row. (4 pts)

Answer: Gene: the recipe for the phenylalanine-breaking enzyme. Enzyme: the worker that breaks down phenylalanine. Job: keep phenylalanine from building up. Typo: the enzyme is misshapen and doesn't work, so phenylalanine piles up and harms the brain. 1 pt per row.

Part B — Decode the recipe

A pretend code, three DNA letters per word: ATG = P, CCA = K, GGT = U, TTA = D, GAC = I, CTG = E, AAT = T. A word not in the table means the cell can't read it — a broken recipe.

2. (a) Decode: ATG CCA GGT. (b) Decode: TTA GAC CTG AAT — what does that recipe spell, and why is it the cure? (c) Now one letter is changed: ATG CCA GCT. What happens when the cell tries to read it? (3 pts)

Answer: (a) PKU (1 pt). (b) DIET — the special low-phenylalanine diet that prevents the harm (1 pt). (c) GCT isn't in the table: one wrong letter (a mutation) breaks the word, so the recipe can't be read — a broken enzyme (1 pt).

Part C — Read the results

3. Four babies' newborn screens came back. For each, write what the result means and what should happen next. Remember: a positive SCREEN means 'test again, properly, now' — not a diagnosis. (1 pt per row for meaning + next step; 2 pts for the last question.) (6 pts)

Answer: A: everything normal — nothing to do; results to the family. B: a positive screen for PKU — repeat the test right away with a proper blood test; if confirmed, start the diet immediately. C: possible congenital hypothyroidism — confirm with a blood test; if confirmed, start the daily thyroid pill. D: a mismatch between hand and foot — blood may be skipping the lungs or a pipe is missing; get an ultrasound of the heart (echo) today. 1 pt per row; 2 extra pts (below).

4. Which of the four babies needs action FIRST, and why? (2 pts)

Answer: Baby D — a possible heart problem can become dangerous within hours or days (the ductus may be the only thing keeping blood flowing), so the echo happens today (1 pt). B and C are urgent too, but the pile-up takes days to weeks to harm, so a repeat test within days is the right speed (1 pt).

Part D — Sort the conditions

5. Draw a line from each condition to what's wrong. (4 pts)

Answer: PKU -> A broken enzyme lets phenylalanine pile up | Congenital hypothyroidism -> The thyroid makes too little of its hormone | Sickle cell disease -> One wrong letter in the hemoglobin recipe; red cells bend | Galactosemia -> An enzyme that can't break down milk sugar (1 pt each)

Part E — Flashbacks

6. From NICU Tutor: the pulse ox is used as a heart screen at 24 hours. Where are the two probes placed, and what is the machine looking for? (1 pt)

Answer: Right hand and a foot. A low reading, or a mismatch between them, suggests blood isn't getting to the lungs properly — a heart built with a missing or blocked pipe.

7. From last week's Genetics: cells read DNA three letters at a time. How many different three-letter words are possible? (1 pt)

Answer: 4 × 4 × 4 = 64.

Part F — Think like a doctor

8. A family gets a phone call: their baby's heel-prick screen came back positive for PKU. The father asks, "So our baby has PKU?" Answer him honestly, using what you learned about screens as nets — and say what you'd do this week. (3 pts)

Answer: Not yet known: a screen is a wide net set to never miss a real case, so it also catches some healthy babies; most positive screens turn out normal on the proper repeat test (1 pt). But we act as if it could be real: repeat the test within days, and if it's confirmed, start the low-phenylalanine diet right away — started early, the baby grows up healthy (1 pt). Be kind and fast: the scary call is the price of a net that doesn't miss, and the family deserves to hear that clearly (1 pt).

HOME LAB (optional, not graded): A working enzyme and a broken one

You need: a raw potato, a cooked piece of the same potato (boiled or microwaved, then cooled), hydrogen peroxide from the first-aid kit, two small cups, a spoon, a grown-up

1. Put a slice of RAW potato in one cup and a slice of COOKED potato in the other.
2. Pour a spoonful of hydrogen peroxide over each slice. Watch for a minute.
3. Which cup fizzes? Feel the fizzing cup — is it warm? What do you think the bubbles are?
4. Write down what each cup did, and which one is 'PKU'.

What you should see: The raw slice foams: an enzyme called catalase snips hydrogen peroxide into water and oxygen gas (the bubbles), and the cup warms a little. The cooked slice does nothing — heat warped the enzyme's shape, so the job doesn't get done and the peroxide just sits there. The cooked potato is 'PKU': same ingredients, broken worker, pile-up.